Enhancing Digital Pathology with Synthetic Data: A Deep-Dive Comparison of cGAN and Diffusion Approaches for Multi-Class Histopathology Generation

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Abstract

Deep learning applications in digital pathology often struggle with data scarcity. Synthetic image generation offers a promising solution. However, researchers rarely compare different types of generative models directly against each other on large, multi-class histopathology datasets. Synthetic image generation can help fill these gaps. In addition, we measured the performance on the downstream task of classification by adding varying amounts of synthetic NORM, MUC and STR patches to the training data set.

In response to this, we compare a conditional GAN (cGAN) with two diffusion models (the DDPM and the latent diffusion model (LDM) with 1000 steps) both trained on NCT-CRC-HE-100K.We evaluate the quality of generated images using the Kernel Inception Distance (KID). We further evaluate the effectiveness of the synthetic images for the training of a downstream model by increasing the number of synthetic NORM, MUC and STR patches in the training data of a ResNet50. Additionally, we measure the training time and inference time per image using a single NVIDIA P100. More importantly, we tested if these synthetic images actually helped a downstream classifier.

You could see a distinct trade-off in the results when it came to image realism versus what is of practical use. On the one hand, the DDPM produced the most lifelike patches by some margin, with a KID of 0.017 for the NORM class. But it was the LDM that turned in the best classification score at 96.70%. In fact, the LDM's images were a bit rougher around the edges and this served as an effective regularizer for the ResNet-50, enabling it to make sense of the difficult boundary between Stroma and the rest of the tissue.

Then there is the matter of hardware. The cGAN was hands down the quickest, churning out 781.38 images a second. That is many times over what you would get from either of the diffusion methods, and even so it still brought the classifier's accuracy to a very respectable 96.50%. It goes to show that you do not have to have pixel-perfect synthetic data to put some improvement into a pathology classifier. What counts is the structural variance you put in. Research teams would be wise to weigh their generative approach against the realities of their computing power.

1 Introduction and Hypothesis of the Study

1.1 Background

Cancer diagnosis and research now increasingly rely on digital pathology, which enables large-scale analysis of histopathology images using deep learning techniques. [1, 2]. Recent advances in convolutional neural networks have demonstrated strong performance in tasks such as cancer detection, grading, and subtype classification [3]. However, the need for large, balanced, and diverse annotated datasets plays a crucial role in the effectiveness of these models. Histopathology datasets are often limited, especially for rare cancer types, which can negatively affect model performance, particularly generalization. [4, 5]. A promising solution to address data limitaion in medical imaging is synthetic data generation. Generative adversarial networks (GANs) have been widely explored to generate and augment histopathology images, improving tasks such as classification and stain normalization. [6–8]. More recently, diffusion-based generative models have gained attention for their stable training dynamics, ability to generate high-fidelity images, and scalability across complex datasets [9, 10]. With promising outcomes, preliminary research has started examining diffusion models in medical imaging, including histopathology. [11]

1.2 Problem Statement

Despite these advancements, much of the existing research focuses on certain tissue types or specific generative model architectures, which makes it difficult to draw general conclusions about the relative performance of GANs and diffusion models [12, 13]. Moreover, only a limited number of studies consider the computational aspects of these methods—such as training duration and inference speed—alongside their impact on downstream classification. Specifically, most existing studies evaluate generative models on a single monolithic tissue type, failing to systematically investigate whether a single generative architecture can consistently synthesize morphologically distinct tissue classes within the same dataset with equal fidelity.

1.3 Objectives

In this paper, we will compare GAN and diffusion-based image generation methodologies for multi-class histopathology image synthesis within a highly diverse, publicly available dataset. Specifically, this study seeks to answer: (1) What is the relative performance of conditional GAN (cGAN) and diffusion models in terms of synthetic image quality across distinct histopathological tissue classes? (2) How does the synthetic data from each approach affect downstream classification performance, particularly for challenging or minority classes? (3) Which generative paradigm is more computationally efficient in terms of training time and inference speed? and (4) Is the generative fidelity of these models consistent across structurally different tissue types within the same dataset?

1.4 Hypothesis and Theoretical Basis

This study is motivated by empirical evidence suggesting that augmenting limited training sets with carefully generated synthetic samples can improve model robustness by increasing data diversity. In data-scarce histopathology settings, such augmentation can reduce overfitting and support more reliable generalization of deep learning models [14]. From this perspective, we hypothesize that diffusion-based generative models will yield more consistent generative quality across diverse tissue classes than GAN-based alternatives. This hypothesis is informed by the likelihood-driven training mechanism of diffusion models, their lower sensitivity to adversarial instability, and their more predictable scaling behavior

2 Related Work

Several studies have explored GAN-based architectures for synthetic histopathology image generation and data augmentation [15,16]. Conditional GANs and style-based GAN variants have been widely adopted to generate realistic tissue patches, improve classification performance, and address data scarcity in small or imbalanced datasets. These approaches have demonstrated notable improvements in downstream tasks such as cancer detection, grading, and virtual staining across a variety of tissue types and datasets [17,18].

However, despite their success, GAN-based methods may exhibit sensitivity to training dynamics and data distribution shifts, particularly when applied to heterogeneous multi-source datasets or limited data regimes [19]. Although modern GAN variants have introduced stabilization techniques and architectural improvements [20], ensuring consistent performance and generalization across diverse histopathology datasets remains a challenge [21,22]. These observations motivate the need for systematic benchmarking of GAN-based approaches against alternative generative paradigms.

In parallel, diffusion-based models have recently emerged as a promising alternative for synthetic histopathology image generation [10]. These models, including Denoising Diffusion Probabilistic Models (DDPMs) and their variants, generate high-fidelity tissue patches by progressively adding noise to images during training and learning to reverse this noising process during generation [11]. This iterative denoising enables the model to capture complex morphological structures more reliably than GANs in certain scenarios [22]. Diffusion models have been applied to improve classification performance [23], enhance dataset diversity, and address data scarcity in small or imbalanced datasets [21]. Compared to GANs, diffusion models often achieve higher fidelity and greater diversity in generated images, demonstrating their ability to better capture complex data distributions across heterogeneous multi-source datasets [22]. However, their computational cost and longer inference times can limit scalability, and systematic evaluations across different tissue types are still limited [24].

Table 1: Summary of Synthetic Data Generation Methods in Histopathology (Newest to Oldest)

Reference	**Method / Model**	**Key Findings**	**Dataset**	**Data Size**	**Accuracy Metric**	**Accuracy / Result**
Rodriguez-Alonso et al. (2025)	GAN, ResNet-18	Improves melanoma detection in low-data regimes.	NIH, PMC	13,000	Accuracy	96.00%

Reference	Method / Model	Key Findings	Dataset	Data Size	Accuracy Metric	Accuracy / Result
Saad et al. (2025)	IHC-GAN (Pix2PixHD-based)	Dual generator module for H&E to IHC translation.	HER2/BCI Dataset	4,870 pairs	FID, PSNR, MSE	FID=0.0927; PSNR=22.87; MSE=631.52
VISGAB et al. (2025)	CycleGAN, CUTGAN, DCLGAN	Virtual staining enhances skin tissue visualization.	E-Staining DermaRepo	87 paired WSIs	HSFI	0.81
Shaban et al. (2025)	cGAN	Effective generation of lymphocyte density maps.	NCT-CRC-HE-100K	100,000 image patches	FID	21.20
Lee et al. (2025)	Med-CDCGAN	Generates diverse and realistic prostate biopsies.	PANDA Prostate	4,279 augmented images	FID, PSNR	FID=1.3; PSNR=10.64 dB
Zhang et al. (2025)	DCGAN + ResNet	Augmentation reduces overfitting in small sets.	Kaggle Breast Cancer	277,524 patches	Accuracy	90.50%
Liu et al. (2025)	FSCA-GAN	Hybrid GAN improves spatial fidelity.	PCAM 200	2,000 images (1k Tumor, 1k Normal)	FID, IS, KID	FID=41.83; IS=2.01; KID=0.015
Kim et al. (2025)	LDM	Better synthesis of complex tissue morphology.	CAM16, PANDA	2k (CAM16); 514k (TCGA)	IoU, FD	PANDA: 0.95 (IoU), 22.8 (FD)
Ruiz-Casado et al. (2024)	BEGAN, ProGAN, RcGAN	Synthetic data boosts classification performance.	Breast cancer histopathology	1820 ben. + 1232 mal.	FID	222.81 (Mal); 261.00 (Ben)
Pozzi et al. (2024)	DDPM, Classifier-free guidance	High realism; synthetic images are diagnostically useful.	GTEx WSIs (Brain, Lung, etc.)	648 WSIs, 62,299 tiles	IS, FID, Acc	IS=2.91; FID=17.15; ACC=0.974
Ramamoorthy et al. (2024)	SRGAN, PFE-INC-RES	Super-resolution improves diagnostic precision.	BreakHis, IDC	7,909	Accuracy	99.84%
Bentaieb et al. (2024)	Proposed GAN	Synthesized images improve biopsy grading accuracy.	PANDA	Not available	Gloss, Dloss	0.72; 0.082
Chen et al. (2024)	MxDiffusion	Synthetic data performs comparably to real data training.	Canine PWT	Small: 1,000; Large: 17,000	FID, Prec/Rec	Small: 118.28; Large: 109.91
Wolleb et al. (2024)	Latent Diffusion Model (LDM)	Synthetic images are indistinguishable for experts.	Patch Camelyon (PCam)	216,868 patches	FID, AUC	FID=90.2, AUC=0.85

Reference	Method / Model	Key Findings	Dataset	Data Size	Accuracy Metric	Accuracy / Result
Wang et al. (2024)	Latent Diffusion Model (LDM)	Precise text-guided nuclei generation.	PanNuke, Lizard	13k, 7.9k, 1.7k patches	FID, IS	PanNuke: 50.29 (FID), 3.59 (IS)
Patel et al. (2024)	DDPM	Synthetic sets as effective as real for model training.	PatchCamelyon (PCam)	327,680 color images	FID, F1-score	FID: 35.70; F1: 17.90
Chen et al. (2023)	Diffusion	Morphology-aware models yield superior structures.	TCIA	33.8K patches from 344 slides	IS, FID, sFID	IS=2.08; FID=20.11; sFID=6.32
Karras et al. (2023)	StyleGAN2-ADA	High-quality synthesis with specific adaptive tuning.	TCGA-COAD	74,000 image patches	FID	6.02 (COAD); 3.03 (PRAD)
Singh et al. (2022)	DNN, cGAN	Useful for explainability and clinical education.	TCGA, CP-TAC	2,358 images	AUROC	0.94 ± 0.03
Zhang et al. (2020)	HistoGAN	Targeted augmentation improves tissue classifiers.	Cervical, PCam	2,647 (Cerv); 32,768 (PCam)	Perf. gain	+6.7% (Cervical); +2.8% (PCam)
Quiros et al. (2019)	BigGAN	Learns meaningful deep tissue representations.	NKI, VGH	310K (after processing)	FID, 1-NN	FID: 18.4 (NKI); 8.21 (VGH)

3 aim of study

3.1 Research Problem Statements:

Problem 1: **Lack of multi-class benchmarking:** Lack of intra-dataset cross-class benchmarking: Most existing studies evaluate generative models using monolithic tissue samples, failing to systematically investigate whether a single generative architecture can consistently synthesize morphologically distinct tissue classes within the same dataset. This limits the generalizability, robustness, and reproducibility of reported results across diverse cellular structures. [19, 25].

Problem 2: **Limited systematic comparison between GAN and diffusion model variations:** Prior works typically focus on either GAN-based or diffusion-based approaches in isolation, with few studies conducting comprehensive and fair comparisons across multiple architectural variants from both paradigms [26, 27].

Problem 3: **Insufficient evaluation under class imbalance conditions:** Although synthetic data is often proposed as a solution for data scarcity, limited attention has been given to analyzing its impact on downstream classification performance for underrepresented classes in imbalanced histopathology datasets [15].

Problem 4: **Lack of computational complexity and efficiency analysis:** Comparative evaluations of training cost, inference time, and computational efficiency between GANs and diffusion models are scarce, limiting their practical adoption in real-world clinical and research settings [23, 28].

3.2 Aims and Objectives of the Study:

To address the aforementioned research gaps, this study aims to provide a unified and systematic evaluation of generative models for synthetic histopathology image generation. The goals are intended to directly close the gaps found and make it possible to properly test the hypothesis.

Aim 1: To increase the robustness of generative benchmarking by analyzing synthetic fidelity across distinctly different histopathological tissue classes within a unified dataset framework.

Aim 2: To evaluate the overall generative performance and image quality of the selected GAN-based and diffusion-based model variations.

Aim 3: Examine the effects of synthetic data augmentation on the performance of histopathology image classification, especially in datasets with significant class imbalance.

Aim 4: To analyze and compare the computational efficiency of GAN and diffusion models, considering training time and resource usage.

Aim 5: To examine the consistency of generative model performance when synthesizing morphologically diverse tissue types, identifying if certain paradigms struggle with specific cellular structures.

3.3 Originality and Contribution:

The novelty of this work is that we propose a unified framework that can simultaneously accommodate multi-class benchmarking, cross-paradigm comparison between GANs and diffusion models, downstream classification evaluation under class imbalance, and computational efficiency analysis. Unlike previous work that is restricted to single class or isolated generative architectures, this paper proposes a comprehensive, reproducible and systematic methodology for evaluating synthetic data generation in digital pathology. The findings provide actionable guidance for the selection of suitable generative models for the augmentation of histopathology datasets, alleviating class imbalance and improving classification performance.

4 Materials and Methods

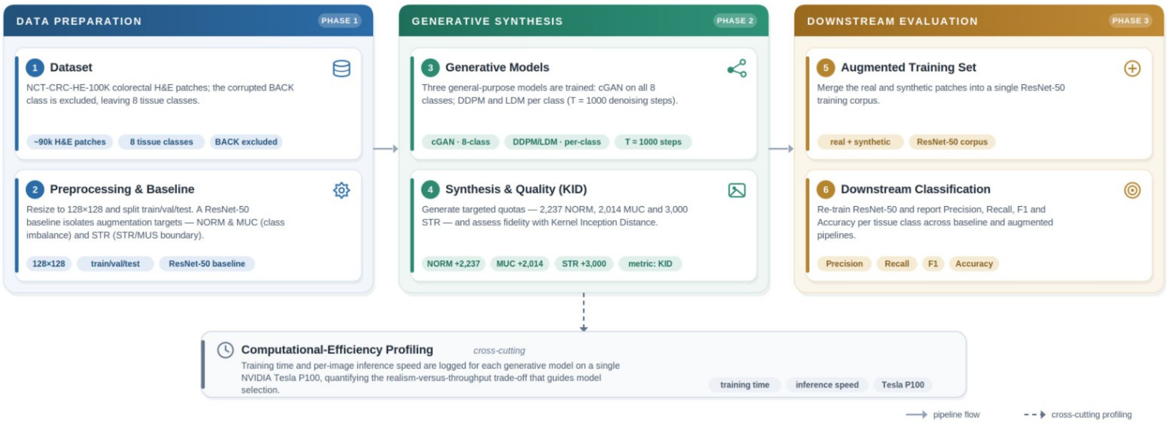


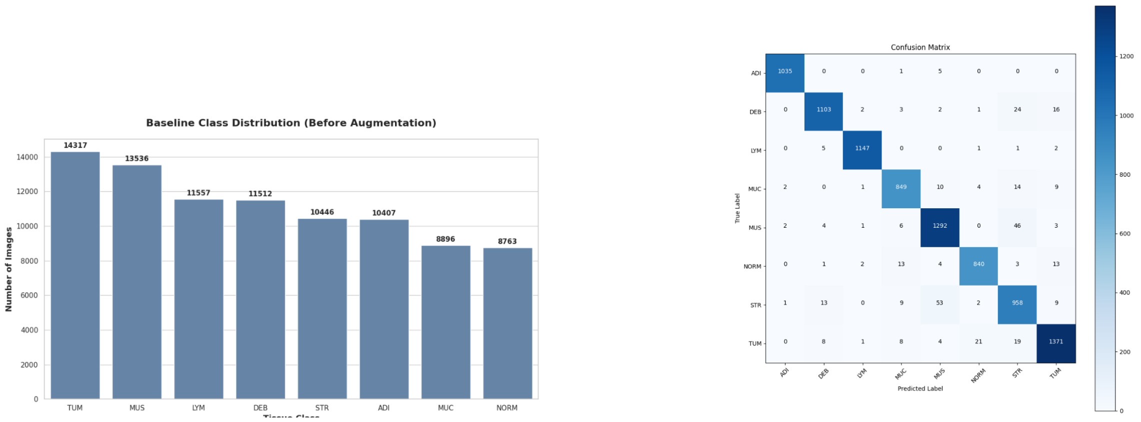
Figure 1: Overview of the workflow for comparing generative models based on accuracy enhancement, evaluation metrics, and time complexity.

4.1 Datasets

This study utilizes the NCT-CRC-HE-100K colorectal cancer dataset. After excluding the corrupted Background (BACK) class due to data integrity issues, an empirical baseline was established using an industry-standard ResNet-50 classifier. Based on quantitative insights from this baseline, three specific tissue classes were isolated for targeted generative augmentation:

- **Data Scarcity Mitigation (NORM & MUC):** As the dataset's smallest minority classes (Figure 2a), NORM and MUC require synthetic expansion to resolve severe class imbalance against dominant classes like Tumor (TUM).

- **Decision Boundary Refinement (STR):** Baseline metrics (Figure 2b) revealed critical morphological confusion, notably misclassifying true Stroma (STR) as Smooth Muscle (MUS). Synthetic variants are required to teach the classifier how to resolve these ambiguous decision boundaries.



(a) Baseline Class Distribution (b) ResNet-50 Confusion Matrix

Figure 2: Baseline dataset metrics prior to generative augmentation. (a) Highlights the severe class imbalance in NORM and MUC, while (b) demonstrates the structural misclassification vulnerabilities within the STR class.

4.2 Methods

Generative models used in this work were purposely selected from general purpose image generation frameworks, rather than domain-specific histopathology models. This was done to ensure that the comparative analysis captures fundamental differences between generative paradigms rather than gains in performance by handcrafted, domain-specific design choices. The aim of the study is to assess the intrinsic appropriateness of GAN and diffusion methods for histopathology data, based only on general models and independently from specialized priors, stain-specific heuristics or pathology-driven architectural changes.

By relying on general models, the study aims to evaluate the intrinsic suitability of GAN and diffusion approaches for histopathology data, independent of specialized priors, stain-specific heuristics, or pathology-driven architectural modifications.

4.2.1 Conditional Generative Adversarial Network (cGAN)

We have selected the conditional Generative Adversarial Network (cGAN) as a generic adversarial baseline, because of its general applicability to a wide range of image domains and explicit conditioning mechanism. Different from domain-specific generative models, the cGAN is independent of handcrafted priors in relation to tissue morphology, staining characteristics or histopathological structure. This design choice allows for an unbiased assessment of class-conditioned image generation without the injection of domain-engineered biases. The cGAN framework generalizes the original GAN formulation by conditioning the generator and the discriminator on auxiliary information, e.g., class labels. This gives the following min-max optimization problem:

$$\min_G \max_D \; \mathbb{E}_{x,y\sim p_{\text{data}}(x,y)} \left[\log D(x,y)\right] + \mathbb{E}_{z\sim p(z),\, y\sim p(y)} \left[\log\left(1 - D(G(z,y),y)\right)\right], \tag{1}$$

here x denotes a real image sampled from the data distribution p_{data}, y represents the corresponding conditioning label, and z is a latent noise vector drawn from a predefined prior distribution $p(z)$. The generator G learns to synthesize images conditioned on y, while the discriminator D aims to distinguish between real and synthetic samples under the same conditioning. To stabilize adversarial training, Spectral Normalization was applied across the discriminator network, alongside a one-sided label smoothing factor of 0.1 during loss calculation.

This model serves as a low-complexity, domain-agnostic reference point for analyzing adversarial training behavior and the effectiveness of conditional synthetic data augmentation under class imbalance.

4.2.2 Denoising Diffusion Probabilistic Model (DDPM)

We chose a *general-purpose diffusion baseline* which is the Denoising Diffusion Probabilistic Model (DDPM) since it has a likelihood-based training objective that is domain agnostic to the image domain. It is entirely based on Gaussian noise processes and does not need any prior knowledge of tissue structure, staining properties or histopathology specific features.

The forward diffusion process progressively corrupts data by adding Gaussian noise at each timestep and is defined as:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I\right), \tag{2}$$

here β_t denotes the noise variance schedule and I is the identity matrix. As t increases, the data distribution approaches an isotropic Gaussian.

Training minimizes the noise prediction loss, where the model learns to estimate the injected noise at each timestep:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right], \tag{3}$$

here $\epsilon \sim \mathcal{N}(0, I)$ represents Gaussian noise and $\epsilon_\theta(\cdot)$ is the learned noise predictor. Furthermore, to ensure stable optimization and maintain global image contrast during the denoising process, the pipeline utilized Min-SNR loss weighting ($\gamma = 5.0$) and Offset Noise (strength = 0.1). This formulation enables the evaluation of *domain-agnostic generative robustness* across heterogeneous histopathology datasets.

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4.2.3 Latent Diffusion Model (LDM)

To address the scalability limitations of pixel-space diffusion models while maintaining domain neutrality, a Latent Diffusion Model (LDM) was selected. LDMs perform diffusion in a compressed latent space learned via an autoencoder, without embedding pathology-specific priors or handcrafted inductive biases. Input images ($128 \times 128 \times 3$) are first encoded into a compressed latent representation ($32 \times 32 \times 4$) using a Variational Autoencoder (VAE):

$$z = E(x), \tag{4}$$

here $E(\cdot)$ denotes the encoder network. The diffusion process is then applied in latent space, significantly reducing computational cost.

The training objective in latent space follows the same noise prediction formulation as DDPM:

$$\mathcal{L}_{\mathrm{LDM}} = \mathbb{E}_{z_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(z_t, t)\|^2\right]. \tag{5}$$

Similar to the DDPM, training utilized Min-SNR weighting ($\gamma = 5.0$) and Offset Noise (strength = 0.1) for stability. This approach enables efficient training on high-resolution datasets while preserving the general-purpose nature of the generative process.

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4.2.4 Rationale for Using General-Purpose Models

This study avoids confounding effects caused by pathology-engineered architectures and handcrafted priors by limiting the experimental comparison to *non–domain-specific generative models*. Therefore, rather than dataset-specific inductive biases, observed differences in image quality, augmentation effectiveness, and computational efficiency can be mainly attributed to the underlying generative paradigm. This design decision improves the comparison's interpretability and the findings' generalizability, allowing them to be applied to medical imaging fields other than histopathology.

4.3 Experimental Protocol

The dataset was preprocessed using standard image resizing (128x128 pixels) and partitioned into training, validation, and test subsets. Both diffusion models (DDPM and LDM) were configured with $T = 1000$ denoising steps to ensure high-fidelity image synthesis. Due to the immense computational and memory constraints inherent to diffusion models, training the DDPM and LDM on the full dataset simultaneously was impossible. Instead, they were trained independently on single tissue classes at a time (NORM: 8,763 images; STR: 10,446 images; MUC: 8,896 images), requiring a separate model instance for each tissue type. Conversely, the highly efficient architecture of the cGAN allowed it to be trained on the entire NCT-CRC-HE-100K dataset (all 8 classes, 90,000 images) simultaneously. It simply utilized its class-conditioning label to isolate the target tissues during the generation phase.To address specific class vulnerabilities, targeted synthetic quotas were used rather than flat augmentation ratios. To resolve the issue of structural scarcity, 2,237 NORM images and 2,014 MUC images were synthetically generated, elevating both minority classes to an operational threshold of 11,000 samples. Additionally, 3,000 STR images were synthesized specifically to inject hard-negative morphological variance, which effectively corrected the baseline classifier's confusion at the decision boundary. The impact of this augmentation strategy was evaluated across three distinct pipelines: a baseline (real data only), real data augmented with cGAN patches, and real data augmented with diffusion patches.

4.4 Experimental Setup and Hyperparameters

To ensure reproducibility and fair evaluation across all generative paradigms, the hyperparameters for the conditional Generative Adversarial Network (cGAN), Denoising Diffusion Probabilistic Model (DDPM), Latent Diffusion Model (LDM), and the downstream ResNet-50 classifier were strictly defined based on the executing codebase.

Due to the VRAM constraints of the NVIDIA Tesla P100 GPU, the generative batch sizes were carefully scaled. The cGAN was trained simultaneously on the entire dataset using a batch size of 32 for 100 epochs, utilizing the Adam optimizer with a learning rate of 2×10^{-4} and momentum parameters $\beta_1 = 0.5, \beta_2 = 0.999$. Conversely, both diffusion models (DDPM and LDM) were constrained to a smaller batch size of 16 due to their heavier memory footprints during the $T = 1000$ iterative Markovian denoising steps. Both diffusion pipelines utilized the Adam optimizer with a learning rate of 1×10^{-4} and a gradient clipnorm of 1.0, training for 100 epochs per target class. The LDM specifically utilized a pre-trained VAE to compress the input into a $32 \times 32 \times 4$ latent representation.

For the downstream classification task, the ResNet-50 model utilized a two-phase transfer learning strategy to prevent catastrophic forgetting. Initially, the pre-trained ImageNet base was frozen, and the custom classification head (incorporating a 0.45 dropout rate) was trained using the Adamax optimizer with a learning rate of 1×10^{-3} for 5 epochs. Subsequently, the top 20 layers of the base model were unfrozen and fine-tuned using the Adam optimizer with a reduced learning rate of 1×10^{-5} for an additional 10 epochs. The classifier processed data with a batch size of 128. A comprehensive breakdown of all architecture and training configurations is detailed in Table 2.

Table 2: Hyperparameter configurations for generative models and the downstream classifier.

Hyperparameter	**cGAN**	**DDPM**	**LDM**	**ResNet-50 (Classifier)**
Architecture Base	DCGAN (Conditioned)	U-Net (Attention)	VAE + U-Net (Latent)	ResNet-50 (Pre-trained ImageNet)
Input / Latent Res.	$128 \times 128 \times 3$	$128 \times 128 \times 3$	$32 \times 32 \times 4$ (Latent)	$128 \times 128 \times 3$
Batch Size	32	16	16	128
Learning Rate	2×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-3} (Head) / 1×10^{-5} (Fine-tune)
Optimizer	Adam	Adam	Adam	Adamax (Head) / Adam (Fine-tune)
Optimizer Params	$\beta_1 = 0.5, \beta_2 = 0.999$	Clipnorm = 1.0	Clipnorm = 1.0	Dropout = 0.45
Denoising Steps (T)	N/A	1000	1000	N/A
Advanced Config.	Spectral Normalization	Min-SNR ($\gamma = 5.0$), Offset (0.1)	Min-SNR ($\gamma = 5.0$), Offset (0.1)	Frozen Base (Phase 1)
Training Epochs	100	100 (Per Class)	100 (Per Class)	15 (5 Head + 10 Fine-tune)
Loss Function	Binary Cross-Entropy	Mean Squared Error	Mean Squared Error	Categorical Cross-Entropy

4.5 Statistical Analysis

The quality of synthetic histopathology images will be evaluated by measuring the similarity of real and synthetic image distributions using Kernel Inception Distance (KID). Downstream classification performance will be measured using Accuracy, Precision, Recall and F1-Score, with emphasis on class-wise metrics to measure how well synthetic data patches classifier weaknesses. The computational efficiency in terms of training time and inference speed is computed and logged.

5 Results and Discussion

5.1 Generative Quality and Consistency

To establish the baseline viability of the synthetic data before classification training, the generative fidelity of the conditional GAN (cGAN), Denoising Diffusion Probabilistic Model (DDPM), and Latent Diffusion Model (LDM) was evaluated using the Kernel Inception Distance (KID). Unlike Fréchet Inception Distance (FID), KID provides an unbiased estimator of spatial and morphological similarity, making it highly suitable for evaluating the structurally complex tissue variations within the NCT-CRC-HE-100K dataset.

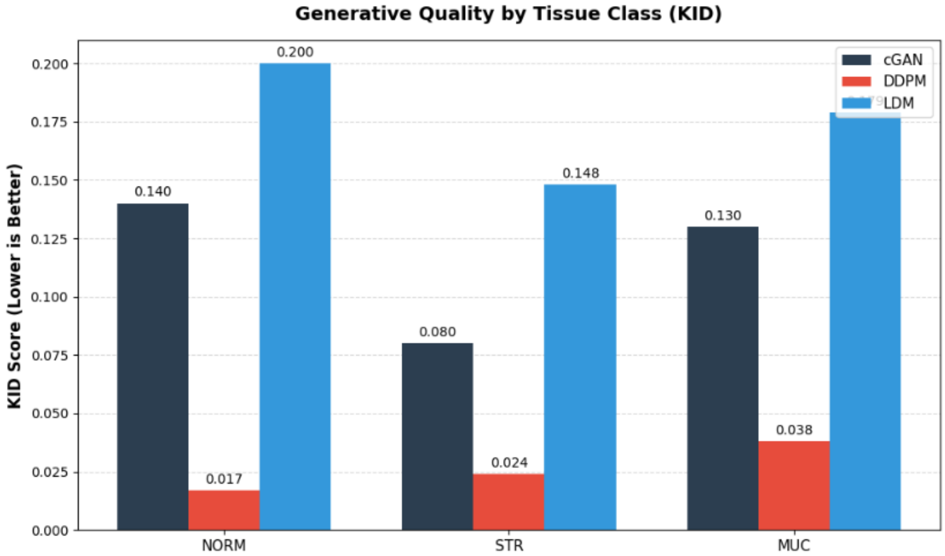


Figure 3: Comparison of the KID across the 3 tissues, Lower KID is better.

As shown in Figure 3, a quantitative assessment of the synthetic outputs shows distinct behaviors of the models. The Denoising Diffusion Probabilistic Model (DDPM) clearly performed the best overall, achieving the lowest KID scores across all three target classes by a significant margin. Additionally, examining consistency across classes highlights some specific structural challenges. While the DDPM achieved excellent morphological fidelity on the Normal Colon Mucosa (NORM) class with a score of 0.017, it showed a slight drop in performance when synthesizing the complex, irregular structures of the Cancer-Associated Stroma (STR: 0.024) and Mucus (MUC: 0.038). In contrast, the cGAN produced higher overall KID scores but interestingly demonstrated its best generative performance on the challenging Stroma class (0.080). For the other classes (NORM: 0.140, MUC: 0.130), the cGAN maintained a tight, consistent performance, showing its stability across different tissue types, even if it could not match the pixel-level realism of the DDPM. Notably, the Latent Diffusion Model (LDM) had difficulty competing in this specific class-isolated training environment, achieving the highest KID scores across the board.

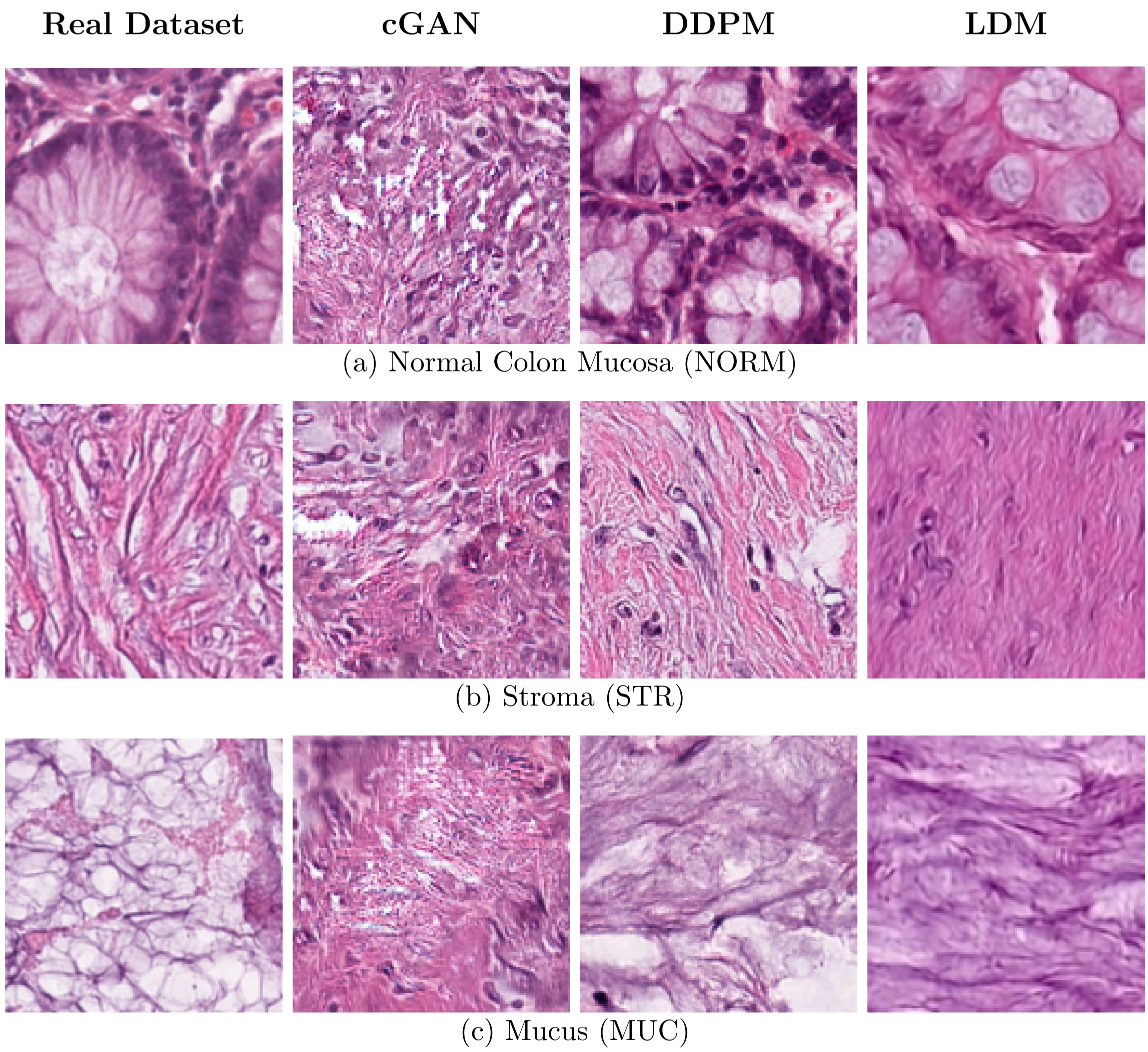


Figure 4: Qualitative visual comparison of synthetic histopathology patches against the real dataset baseline. The matrix demonstrates the morphological fidelity and structural variance captured by each generative paradigm across the three target classes.

5.2 Downstream Classification Performance

We tested the four different classification pipelines set up in Section 4.3 to see if the synthetic augmentation was successful in addressing the structural vulnerabilities of the dataset: (1) a baseline trained only on real data, (2) a classifier augmented with cGAN patches, (3) a classifier augmented with DDPM patches and (4) a classifier augmented with LDM patches. Performance was measured by Precision (P), Recall (R) and F1-score. To explicitly demonstrate the effect of the targeted generative quotas, the metrics in Table 2 are structurally partitioned: the top part isolates the three target classes that received synthetic injections (NORM, MUC and STR), while the bottom part reports the cascading system-wide effects on the unaugmented classes.

Table 3: Comparative classification performance before and after targeted augmentation.

Tissue Class	Baseline (Real Only)			Real + cGAN			Real + DDPM			Real + LDM		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Targeted Augmented Classes												
Normal Mucosa (NORM)	0.97	0.96	0.96	0.97	0.96	**0.97**	0.97	0.97	**0.97**	0.97	0.96	**0.97**
Mucus (MUC)	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96
Stroma (STR)	0.90	0.92	0.91	0.91	0.93	0.92	0.92	0.93	0.92	0.92	**0.93**	**0.93**
Unaugmented Neighboring Classes												
Adipose (ADI)	1.00	0.99	0.99	0.99	1.00	0.99	0.99	1.00	0.99	0.99	0.99	0.99
Debris (DEB)	0.97	0.96	0.97	**0.98**	0.97	0.97	**0.98**	0.96	0.97	**0.98**	0.97	0.97
Lymphocytes (LYM)	0.99	0.99	0.99	1.00	0.99	0.99	0.99	0.99	0.99	0.99	1.00	0.99
Smooth Muscle (MUS)	0.94	0.95	0.95	**0.95**	0.95	0.95	**0.95**	0.95	0.95	**0.95**	0.95	0.95
Tumor (TUM)	0.96	0.96	0.96	0.96	0.97	**0.97**	0.96	0.97	0.96	0.96	0.97	**0.97**
Overall Accuracy	**96.00%**			**96.50%**			**96.60%**			**96.70%**		

As shown in Table 3, the use of targeted synthetic data effectively fixed the baseline's classification issues. This improvement raised the overall diagnostic accuracy across all enhanced pipelines, reaching a high of 96.70% with the LDM.

Importantly, the targeted addition of 3,000 synthetic Stroma (STR) images successfully clarified the unclear decision boundary found in the baseline evaluation. The baseline ResNet-50 was confused in both directions but all three generative pipelines improved the STR F1-score from 0.91 to a maximum of 0.93. Challenging stroma variants were added to the training dataset so that the classifier would create a stricter decision boundary. The boundary correction also had a stabilising effect on the corresponding, unaugmented classes. For example, Smooth Muscle (MUS) and Debris (DEB) which were previously suffering from overlap issues, demonstrated large improvements in precision from 0.94 to 0.95 and 0.97 to 0.98 respectively. This suggests that correcting a specific boundary helps to maintain the accuracy of predictions for neighbouring tissue.

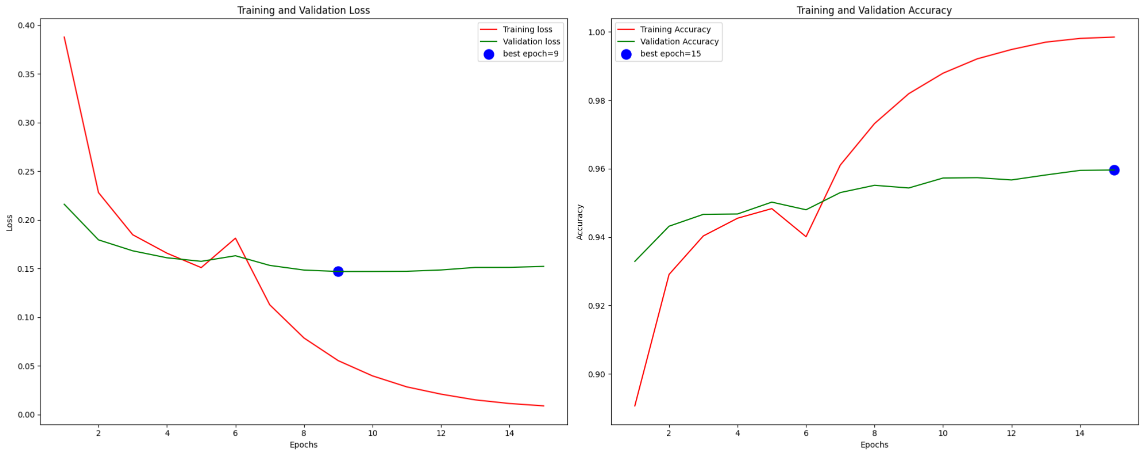
Moreover, owing to the (NORM) class being under-represented, its F1-score improved from 0.96 to 0.97, and the (MUC) class kept consistent metrics without overfitting. Most importantly, the downstream classification performance demonstrated a significant inverse relationship with respect to the quality of the generative data. The LDM that performed the worst in KID in Section 5.1, performed the best downstream with an accuracy of 96.70% and a STR F1-score of 0.93. This points to an important insight: synthetic data does not need to be pixel-perfect to be useful for augmentation. Instead, the higher structural variability added by the LDM acted as a stronger regularizer and encouraged the classifier to learn more robust decision boundaries compared to the high-fidelity but possibly less diverse samples generated by the DDPM.

5.3 Training and Validation Dynamics

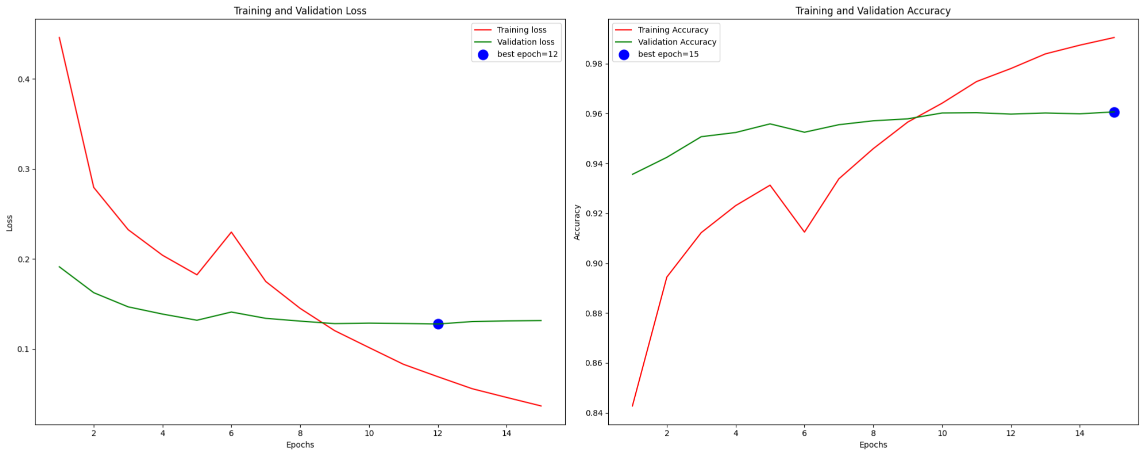
To rigorously assess the impact of the synthetic data on the learning stability of the downstream classifier, training and validation curves (accuracy and loss over epochs) were extracted for the baseline, cGAN-augmented, DDPM-augmented, and LDM-augmented pipelines.

As illustrated in Figure 5, the learning dynamics shift significantly across the different pipelines. The baseline model exhibits classic signs of early overfitting: its validation loss reaches its minimum early at epoch 9 before steadily diverging upward, even as the training accuracy aggressively climbs toward 1.0. In contrast, the injection of synthetic patches generated by the cGAN significantly stabilizes this curve, smoothing the validation trajectory and extending the validation loss minimum to epoch 12.

Meanwhile, the diffusion-augmented pipelines (DDPM and LDM) display more volatile early learning dynamics—marked by a distinct performance dip around epoch 6—but ultimately achieve robust convergence. The DDPM pipeline accelerates validation performance, hitting its optimal loss at epoch 10 and peak accuracy at epoch 11. The LDM pipeline offers a slightly extended learning phase, achieving its lowest validation loss at epoch 12 and its highest validation accuracy at epoch 14. Ultimately, the structural variance introduced by all three generative methods acts as an effective regularizer, delaying the baseline's early overfitting and pushing the classifier to learn more robust decision boundaries.

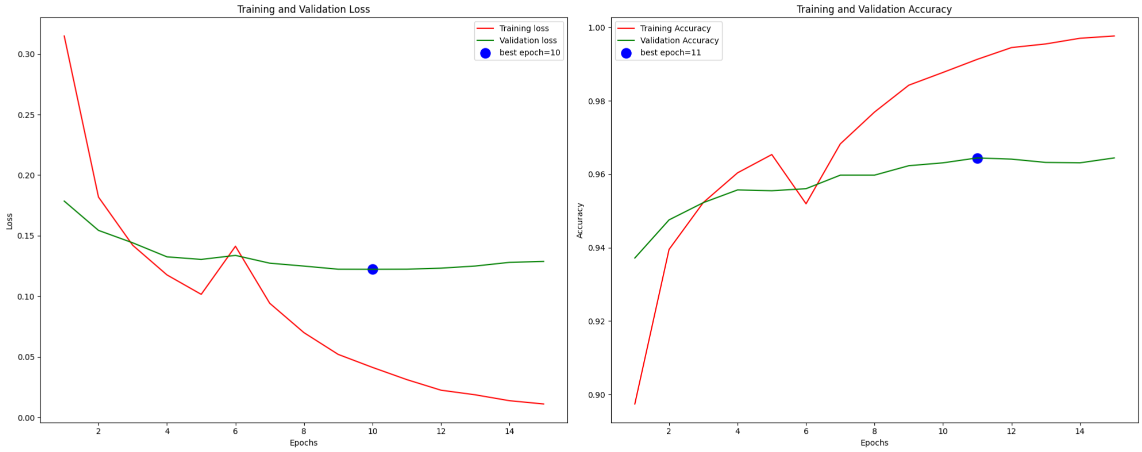


(a) Baseline (Real Data Only)

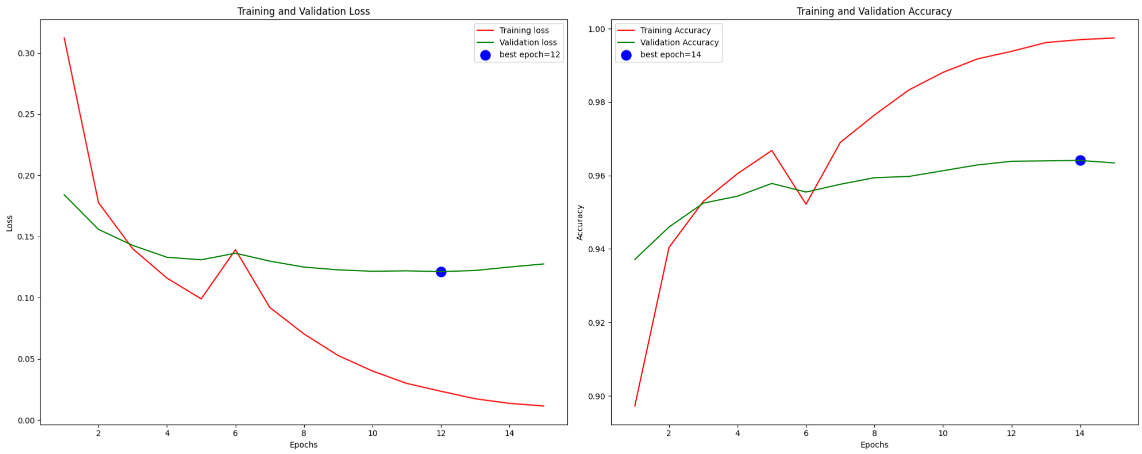


(b) Baseline + cGAN

Figure 5: Training and validation accuracy/loss curves over 15 epochs for the ResNet-50 classifier. (Part 1)



(c) Baseline + DDPM



(d) Baseline + LDM

Figure 5: Training and validation accuracy/loss curves over 15 epochs for the ResNet-50 classifier. (Part 2)

5.4 Computational Efficiency and Resource Trade-offs

A comprehensive assessment of the generative paradigms was given by analyzing the computational complexity, ie., the training duration and data throughput. To provide a uniform ground for comparison, all generative models were trained and evaluated in a single hardware setting with a single GPU (NVIDIA Tesla P100). While Sections 5.1 and 5.2 established the morphological and classification advantages of diffusion-based architectures, these benefits must be contextualized against their inherent computational costs, which often present significant barriers to scalability in clinical deployments.

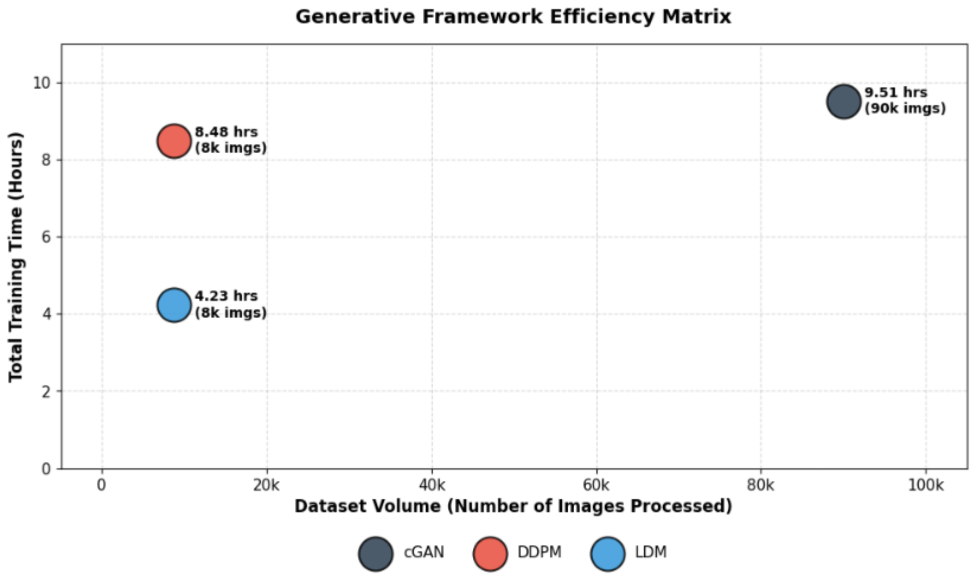


Figure 6: Generative framework efficiency matrix comparing dataset volume to total training duration on a standardized NVIDIA P100 GPU environment.

As illustrated in the efficiency matrix (Figure 6), the conditional GAN (cGAN) demonstrated unparalleled architectural throughput. Capable of training on the entirety of the dataset (approximately 90,000 images across all eight classes) simultaneously, the cGAN completed its training cycle in 9.51 hours on the P100 accelerator. This translates to an exceptionally high processing rate, highlighting the adversarial framework's scalability for massive histopathology datasets that require rapid, on-the-fly augmentation.In stark contrast, the diffusion paradigms incurred severe computational penalties due to their iterative denoising formulations. The Denoising Diffusion Probabilistic Model (DDPM) required 8.48 hours to train on a single isolated class (Normal Mucosa, comprising only 8,763 images). The Latent Diffusion Model (LDM) successfully mitigated a portion of this pixel-space bottleneck via latent compression, cutting the single-class training time down by half to 4.23 hours. However, because the diffusion models required separate, individually trained instances to generate the synthetic quotas for all three target classes (NORM, MUC, and STR), their cumulative computational cost exponentially exceeded the unified cGAN approach.

Ultimately, this presents a critical deployment trade-off. While the LDM achieved the highest downstream classification accuracy (96.70%), its computational overhead remains significantly heavier per image than adversarial methods. For time-sensitive or resource-constrained diagnostic environments, the cGAN provides an optimal balance, delivering highly competitive boundary correction (96.50% accuracy) at a fraction of the hardware cost.

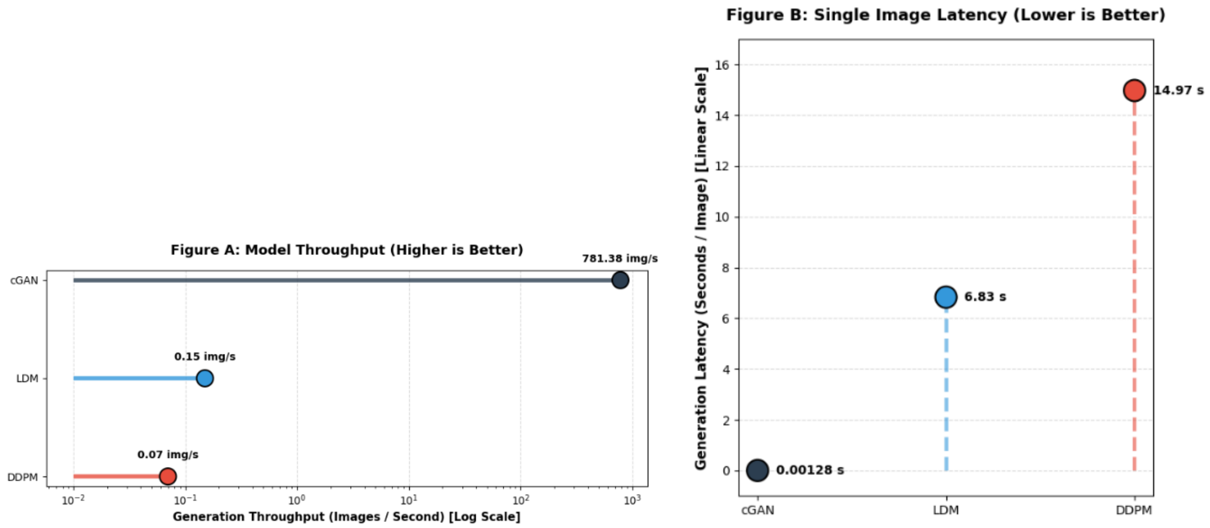


Figure 7: Inference performance benchmarks on a standardized NVIDIA P100 GPU. **Left (A):** A logarithmic representation of generation throughput, highlighting the cGAN's operational dominance. **Right (B):** Single-image latency, demonstrating the severe computational wait times inherent to the diffusion denoising processes.

Beyond the training overhead, the inference overhead for the time it takes to generate a new synthetic histopathology patch highlights the different architectures in both the training and inference paradigms. For a reproducible benchmark, we compared the generation speed of each model on a standard 80 images quota in our NVIDIA Tesla P100 GPU-based environment.

As illustrated in Figure 7, shows that the conditional GAN (cGAN) has an astonishingly fast inference speed. It can be estimated from one forward pass of the generator network that it can compute 781.38 images per second with an average time of 0.00128 seconds per image. This indicates that the adversarial system is an ideal candidate for real-time, scalable and on-the-fly data augmentation in clinical decision support systems.

Unlike the diffusion models, the latter models exhibit high computational latency in generating images due to the iterative Markovian denoising steps. The Denoising Diffusion Probabilistic Model (DDPM) can only produce 0.07 images per second and requires nearly 15 seconds (14.97s) to produce one tissue patch. The Latent Diffusion Model (LDM) can produce 0.15 images per second and requires (6.83s) to generate a single image.

The study demonstrates a crucial trade-off in digital pathology. The LDM achieved the highest downstream accuracy (96.70%) despite having the lowest KID , but it was orders of magnitude slower than adversarial methods in training and inference. For time-sensitive, large-scale, or hardware-limited diagnostics, the cGAN offers a significantly better balance with an accuracy level of (96.50%) but with minimal computational overhead.

6 Conclusion

This study involved the testing of three different generative models- cGAN, DDPM, and LDM- in our effort to find the most effective model that can fix class imbalance and muddy decision boundaries in the NCT-CRC-HE-100K dataset. We were anticipating that diffusion models would be the ones creating the most consistent tissue patches. And the case was indeed as we assumed. The DDPM far surpassed the rest of the models' capability of generating the most realistic images, securing the overall best Kernel Inception Distance (KID) scores.

One of the things we didn't really expect was how those images would actually affect the downstream classifier. There was a marked contrast in visual realism and usefulness for diagnostics. The LDM were actually the ones having the worst KID scores in the group. Nevertheless, because its images were a little bit noisier and incorporated "hard-negative" structural variance, it made the ResNet-50 model to learn better boundaries. The LDM patches resultant from the augmentation of the training set were the driving forces for our peak accuracy rate of 96.70%, and they also fixed the bidirectional confusion in the Stroma class.

Not only precision, but time spent in computing is also crucial. Profiling the hardware exposed massive bottlenecks in the diffusion pipelines, which are the issues that need to be addressed to make pipelines perform optimally. Scaling them up for huge datasets is very tricky since the DDPM and LDM are using step-by-step Markovian denoising. Whereas the cGAN not only caught up but also outperformed them by generating more than 780 images each second over the pipeline. In addition to that, the experiment also confirmed that the cGAN is the most suitable model for large deployment due to the good cGAN performance, which was achieved with a very respectable 96.50% classification accuracy.

The model selection for digital pathology essentially boils down to the hardware and expected outcomes of a lab. If a team owns the hardware resources along with infinite computation time, then the right model for them will be the latent diffusion model to adopt to the class boundaries. But if the primary goal is to have a running high-throughput, real-time clinical pipeline, then the only solution is the adversarial networks along with the cGAN for now. The next major step that researchers should take is to create hybrid architectures. What if we join the quick inference of a GAN with the wide variance latent space of a diffusion model? By doing that, we could get both super high scalability and maximum diagnostic fidelity.

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